



Cloud Computing Introduction

Stefan Schmöller & Lukas Buchner



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Survey:

Do you have experience with the cloud?

What experience do you have?

Who here has already developed software for the cloud?

Who here has already used software that runs in the cloud?

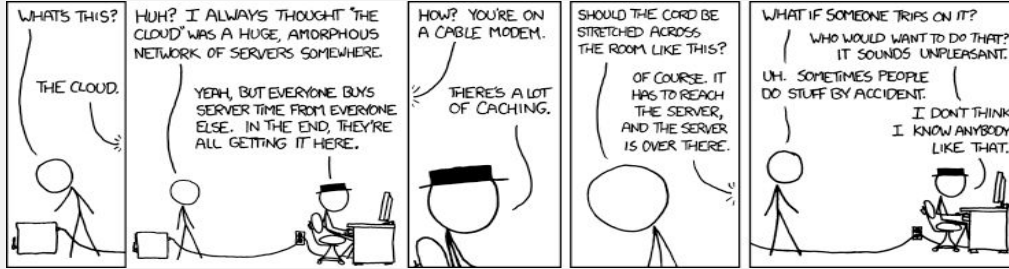
Who here has ever used Azure, AWS, Google Cloud, etc.?



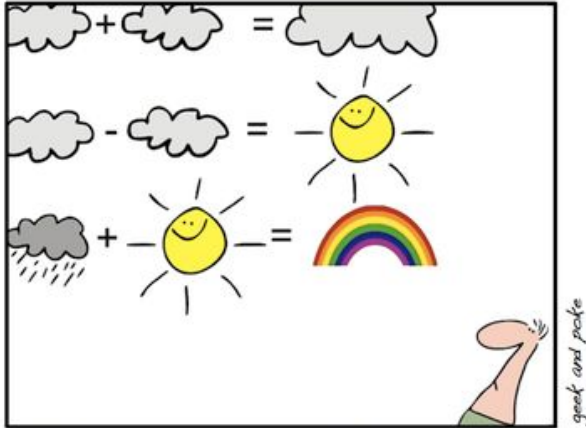
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What is Cloud Computing?

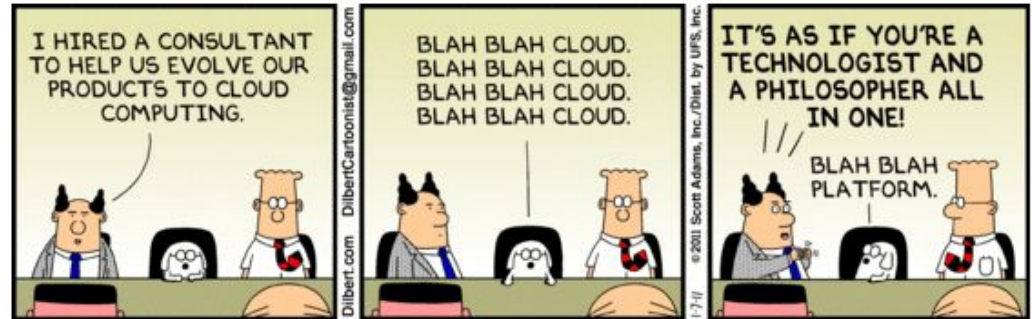
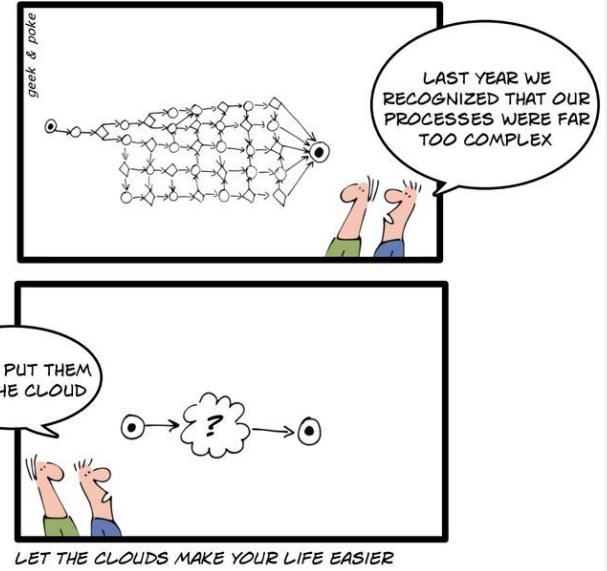
What is Cloud Computing?



There's planned downtime every night when we turn on the Roomba and it runs over the cord.

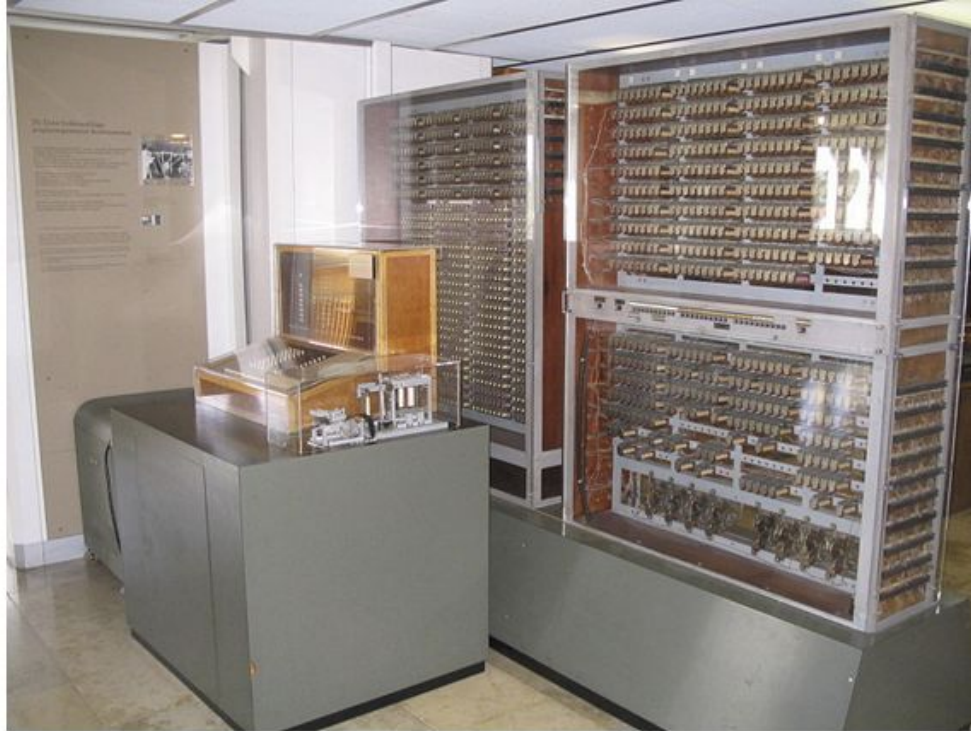


SIMPLY EXPLAINED - PART 17:
CLOUD COMPUTING



Generation 0: The first computers

Quelle: wikipedia.de



Zuse Z3

Rechenleistung: 2 FLOPS

Preis: unbekannt

Generation 1: Mainframes



IBM AS/400

Rechenleistung: 4,5 bis 250 Mega-FLOPS

Preis: 6.000-9.000 \$/Monat

Quelle: wikipedia.de



Cray 1

Rechenleistung: 80 bis 133 Mega-FLOPS

Preis: 5-8 Mio. \$

Generation 2: Supercomputers and Data Centers

Quelle: strato.de



Strato Rechenzentrum:

Rechenleistung: ca. 2,6 Peta-FLOPS

K-Computer (Supercomputer):

Rechenleistung: 10,5 Peta-FLOPS

Computer performance

Name	FLOPS
yottaFLOPS	10^{24}
zettaFLOPS	10^{21}
exaFLOPS	10^{18}
petaFLOPS	10^{15}
teraFLOPS	10^{12}
gigaFLOPS	10^9
megaFLOPS	10^6
kiloFLOPS	10^3

Generation 3: Ubiquitous Computing or „The commoditization of computing power“



Apple A13 (iPhone 11 Pro)

ca. 155 GFlops

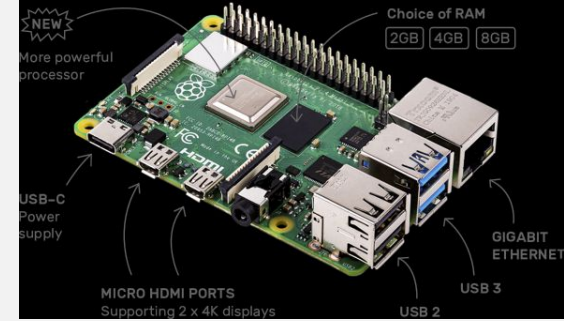
Price: ~1000 €



Nvidia RTX 3090

ca. 36 TFlops

Price: 1.500 €

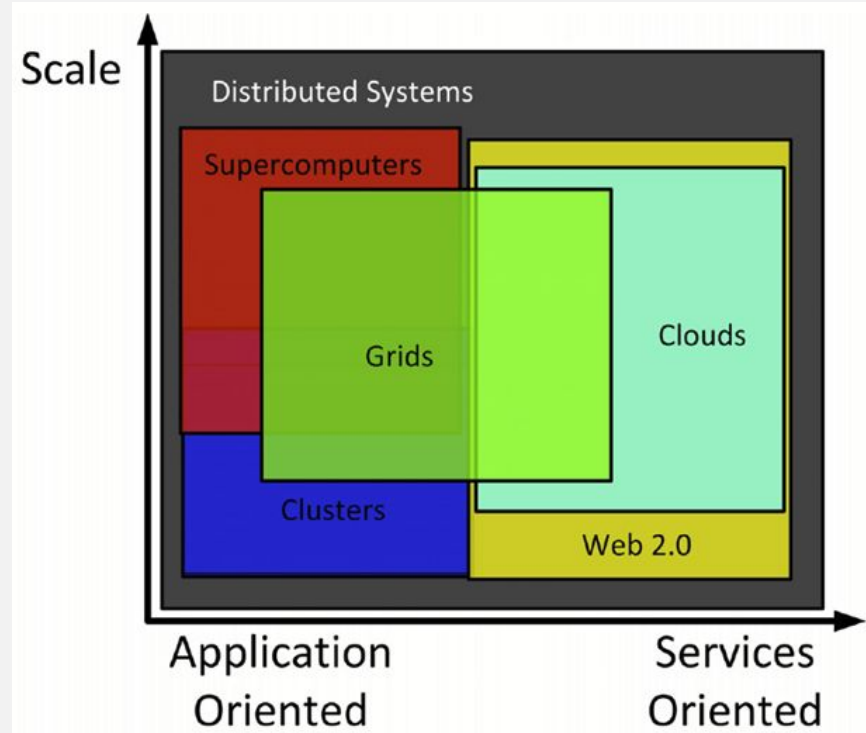


Raspberry Pi 4

ca. 13,5 GFlops

Price: 30 €

Classification of cloud computing in relation to other approaches for distributed systems.



Cloud computing is the result of the commoditization of computing power, computing capacity and the internet.

noun - digital technology: internet-based computing in which large groups of remote servers are networked so as to allow sharing of data-processing tasks, centralized data storage, and online access to computer services or resources.

<http://dictionary.reference.com>

“Cloud computing is the **access to computers and their functionality via the Internet** or a local area network. Users of a cloud request this access from a set of web services that manage **a pool of computing resources** (i.e., machines, network, storage, operating systems, application development environments, application programs). When granted, **a fraction of the resources in the pool is dedicated** to the requesting user until he or she releases them.”

<http://open.eucalyptus.com/learn>

“A large-scale **distributed computing paradigm** that is driven by **economies of scale**, in which a **pool of abstracted, virtualized, dynamically-scalable, managed computing power, storage, platforms, and services** are delivered on **demand** to external customers **over the Internet**.”

Ian Foster et al., Cloud Computing and Grid Computing 360-Degree Compared

The NIST definition of Cloud Computing

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

<https://www.nist.gov/publications/nist-definition-cloud-computing>

Essential properties of cloud computing:

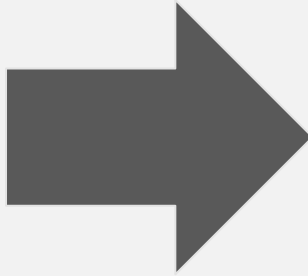
- **On-demand** self-service
No interaction between people!
- Broad **network access**
Internet-based administration
- Resource **pooling**
Multiple customers are using the same resources
- Rapid **elasticity**
Rapid scaling with seemingly unlimited resources
- **Measured** service
Resource utilization measurement (e.g. for billing)

Resources are, for example:

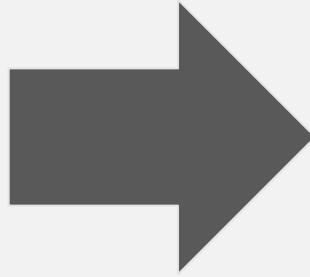
- **Storage**
- **Compute**
- **Memory**
- **Network**
- ...

High number of IT resources

First Cluster



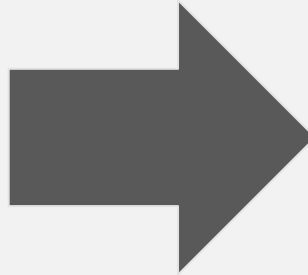
Commodity hardware



High degree of distribution

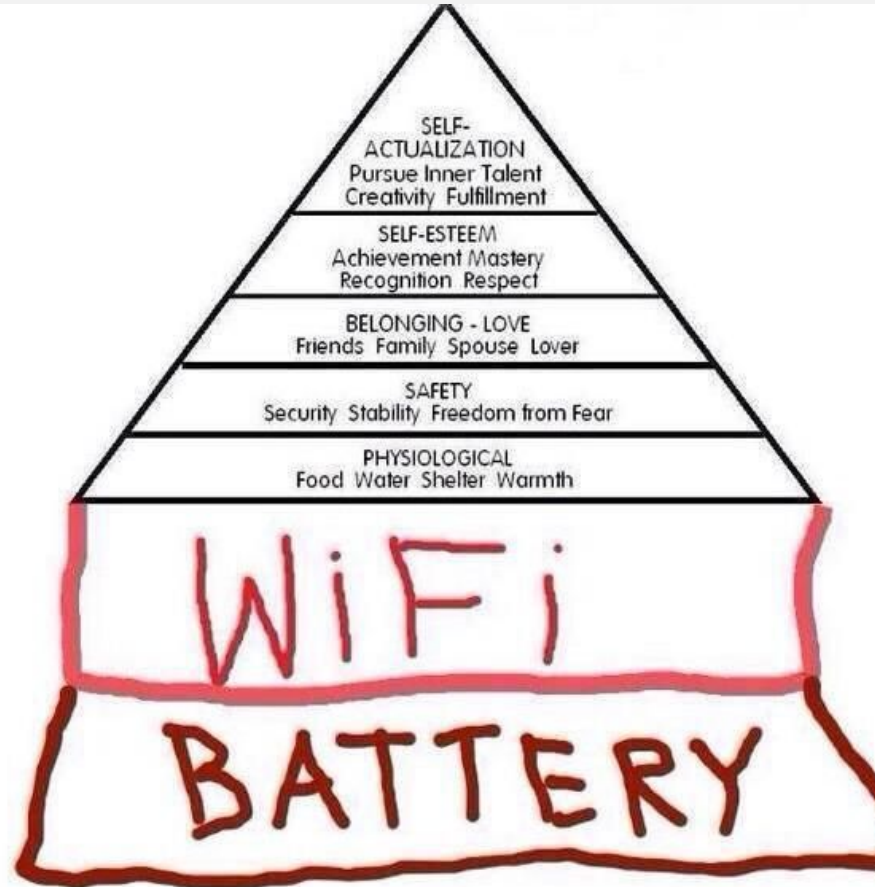


Summit
200 Petaflops



Folding @ home
2,43 Exaflops

The commoditization of the internet.



So what is Cloud Computing?



At its core, cloud computing is about a shallower level of integration in system development and operation.

Applications

Libraries

Software infrastructure

Operating system (OS)

Hardware

IT resources from the cloud
that can be consumed on
demand.

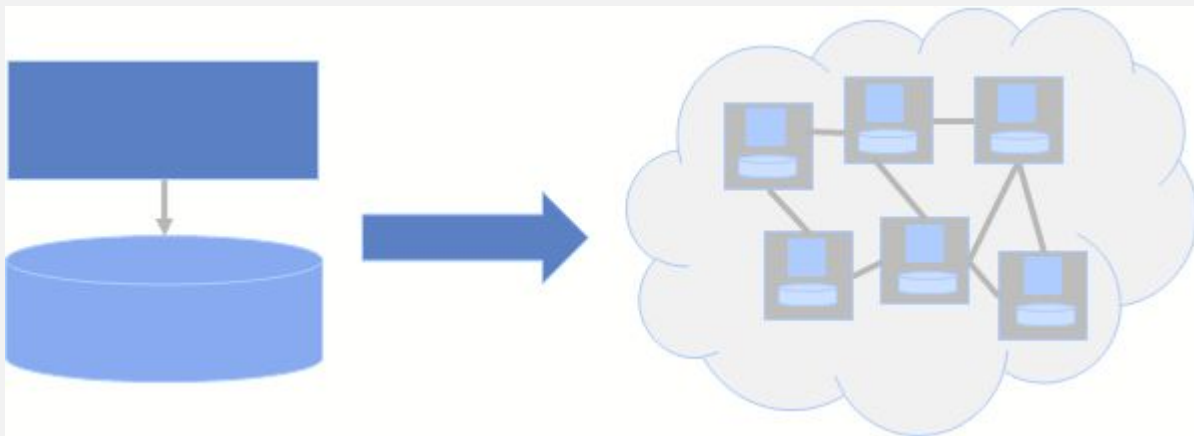


“computation may someday be organized as a public utility”, John McCarthy, 1961

Probably the most important technical influence on how we will build software systems in the future.



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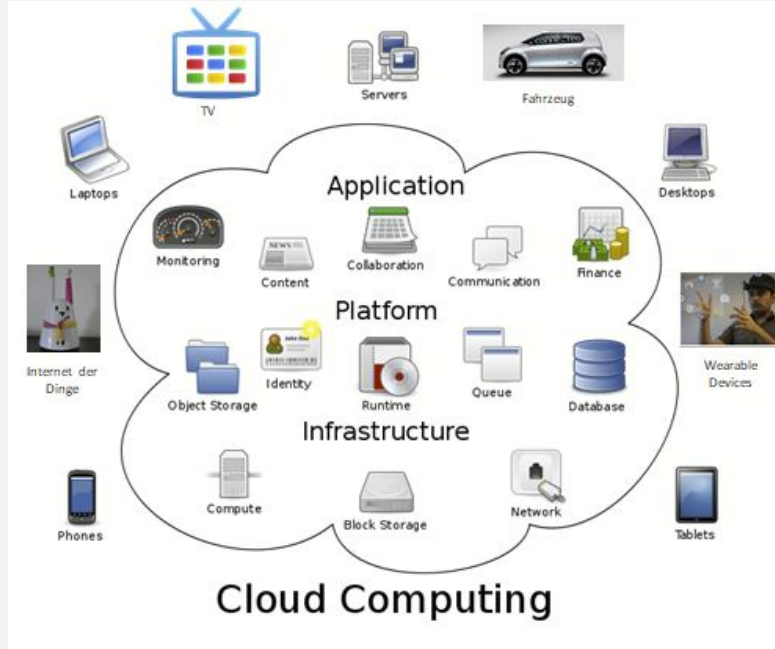


- Distributed data
- Distributed logic
- Distributed consistency
- Diagnosability
- Elasticity
- Provisioning
- Orchestration
- Scheduling
- Service Discovery

The cloud is dynamic, elastic and omnipresent.

The most important properties of Cloud Computing:

- **X as a Service:** On-demand character; provision of computing capacity, platform services and applications on request and in real time.
- **Resource pools:** Availability of seemingly unlimited resources that process requests in a distributed manner.
- **Elasticity:** Dynamic allocation of additional resources as needed (self-adaptation). No more capacity planning necessary from the user's point of view.
- **Pay as you go model:** Economy of Scale. The costs scale with the benefits.
- **Omnipresence:** Access to the cloud via the internet and from a wide range of end devices (via standard protocols).



The 5 Commandments of the Cloud

- Everything fails all the time
- Focus on MTTR, not on MTTF
- Respect the eight fallacies of distributed computing
- Scale out, not up
- Treat resources as cattle, not as pets



Quelle: https://de.wikipedia.org/wiki/Zehn_Gebote

Eight fallacies of distributed computing

DeveloperToArchitect.com

Software Architecture Monday with Mark Richards Lesson 18 - Fallacies of Distributed Computing



Mark Richards

Independent Consultant

Hands-on Software Architect / Published Author / Conference Speaker

Founder, DeveloperToArchitect.com

www.wmrichards.com

Eight fallacies of distributed computing



Cloud benefits

Temporary servers

- Project servers
- Test servers
- Prototype servers

Easy deployment

- Automatic application deployment
- Automatic creation of various deployment variants

Scalable applications

- Dynamic scaling, depending on the request load

Comprehensive calculations

- Transaction analysis
- Data aggregation
- Data warehousing



- Needed offline conversion of public domain articles from 1851-1922.
- Used Hadoop to convert scanned images to PDF
- Ran 100 Amazon EC2 instances for around 24 hours
- 4 TB of input
- 1.5 TB of output

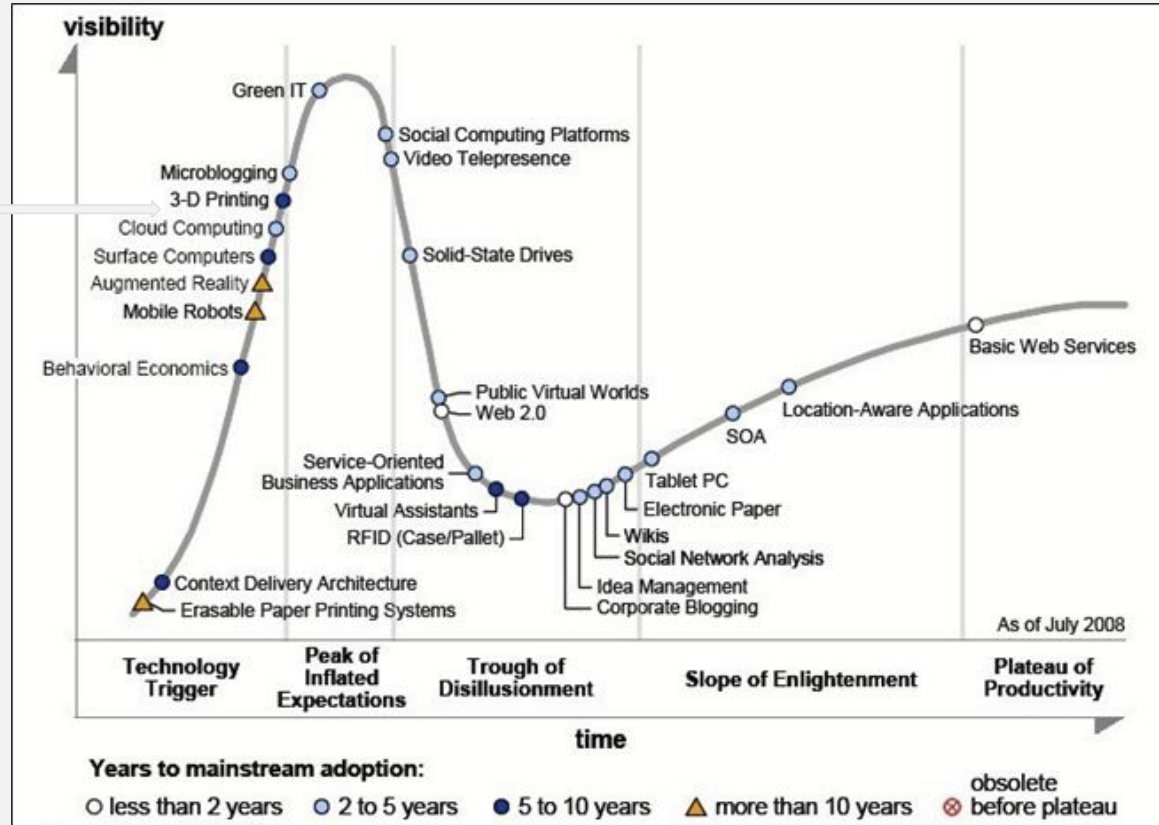
A COMPUTER WANTED.
WASHINGTON, May 1.—A civil service examination will be held May 18 in Washington, and, if necessary, in other cities, to secure eligibles for the position of computer in the Nautical Almanac Office, where two vacancies exist—one at \$1,000, the other at \$1,400. The examination will include the subjects of algebra, geometry, trigonometry, and astronomy. Application blanks may be obtained of the United States Civil Service Commission.

Published 1892, copyright New York Times

<http://www.slideshare.net/acarlos1000/hadoop-basics-presentation>

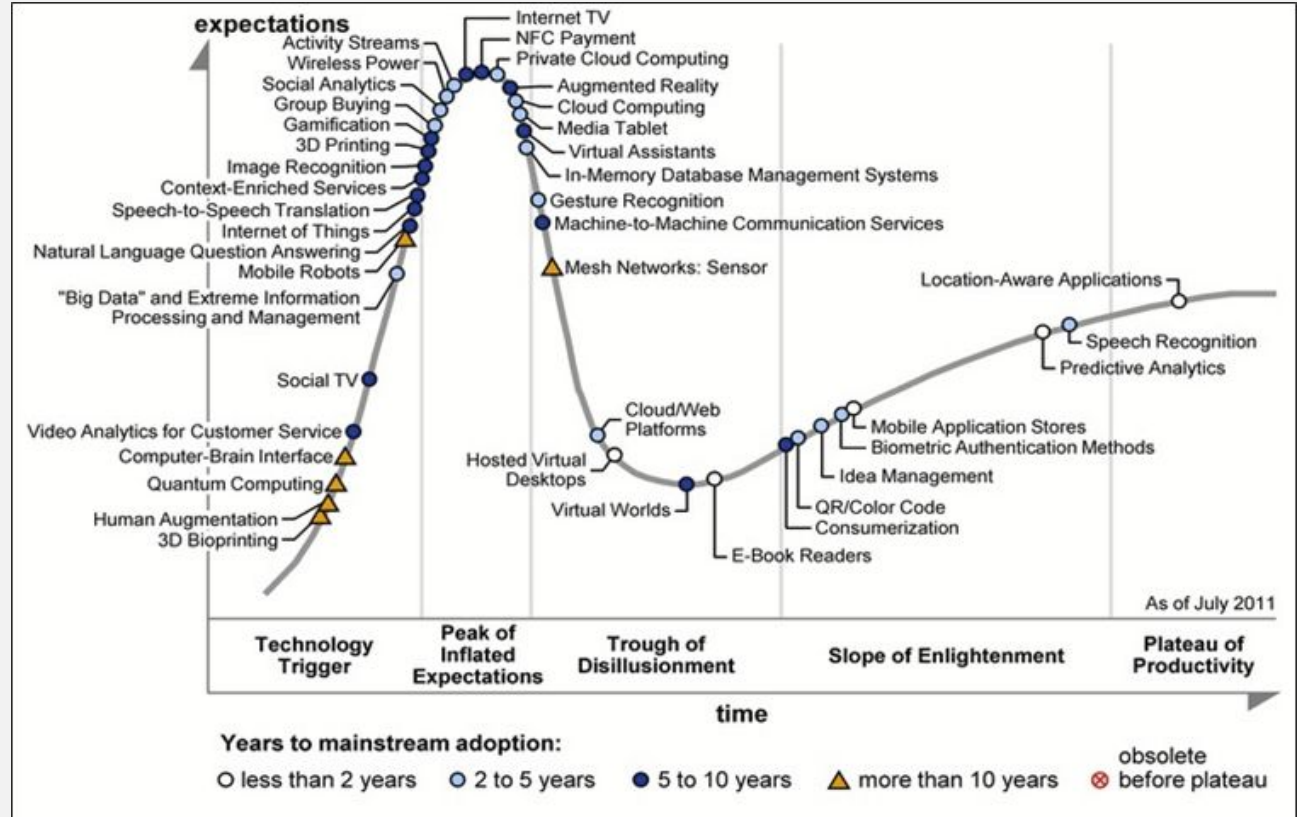
Cloud computing is one of the biggest IT trends of recent years. (2008)

Gartner Emerging Technologies Hype Cycle, 2008



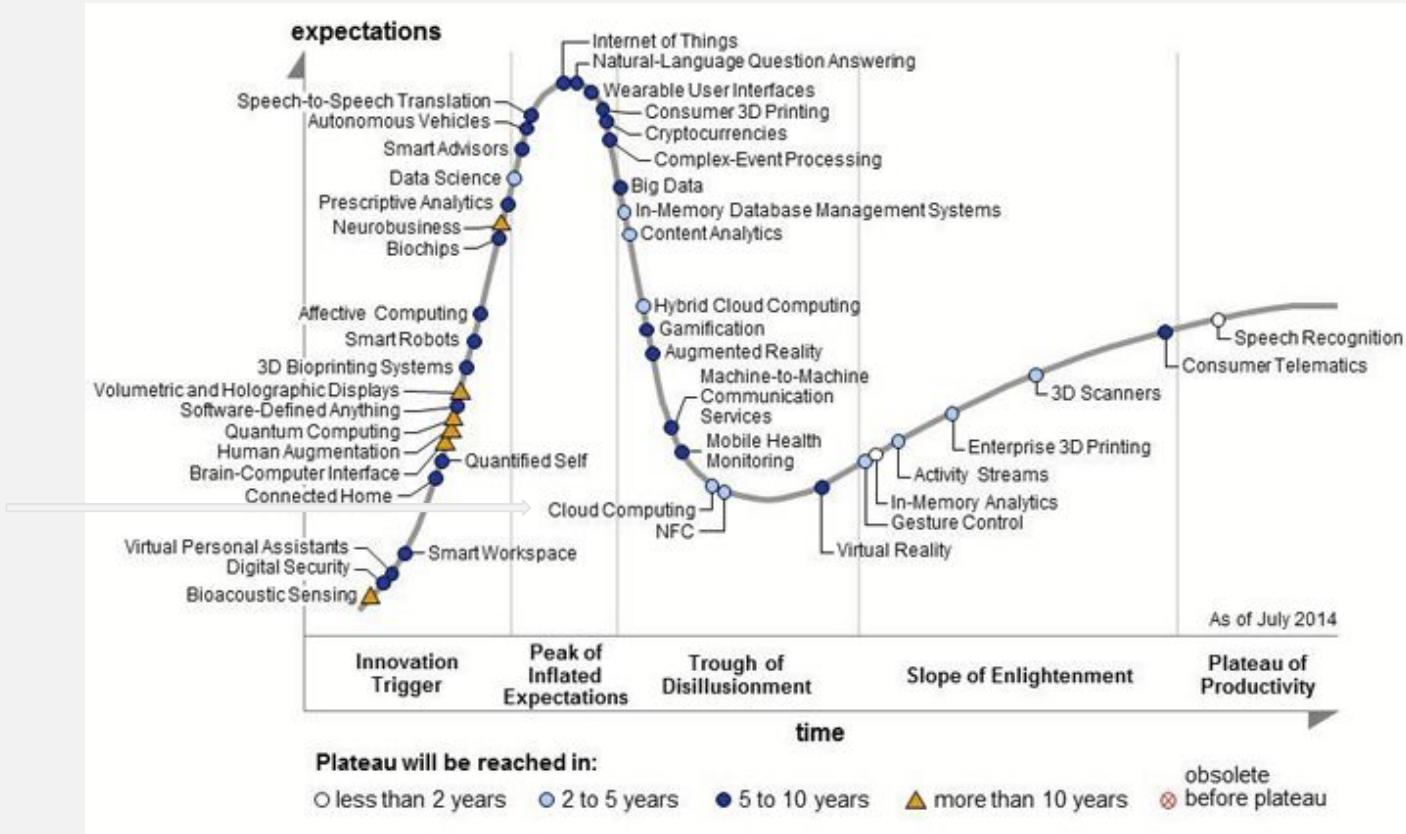
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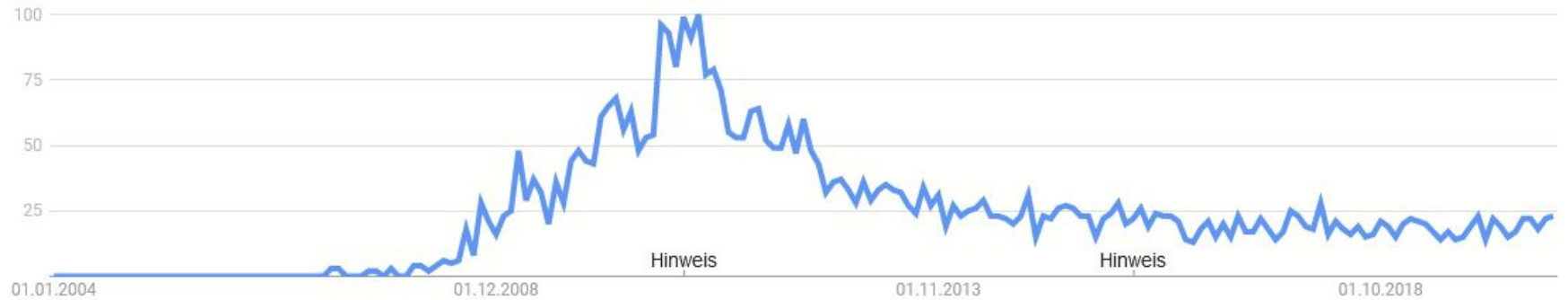
But it returns to the facts. (2014)

Gartner Emerging Technologies Hype Cycle, 2014



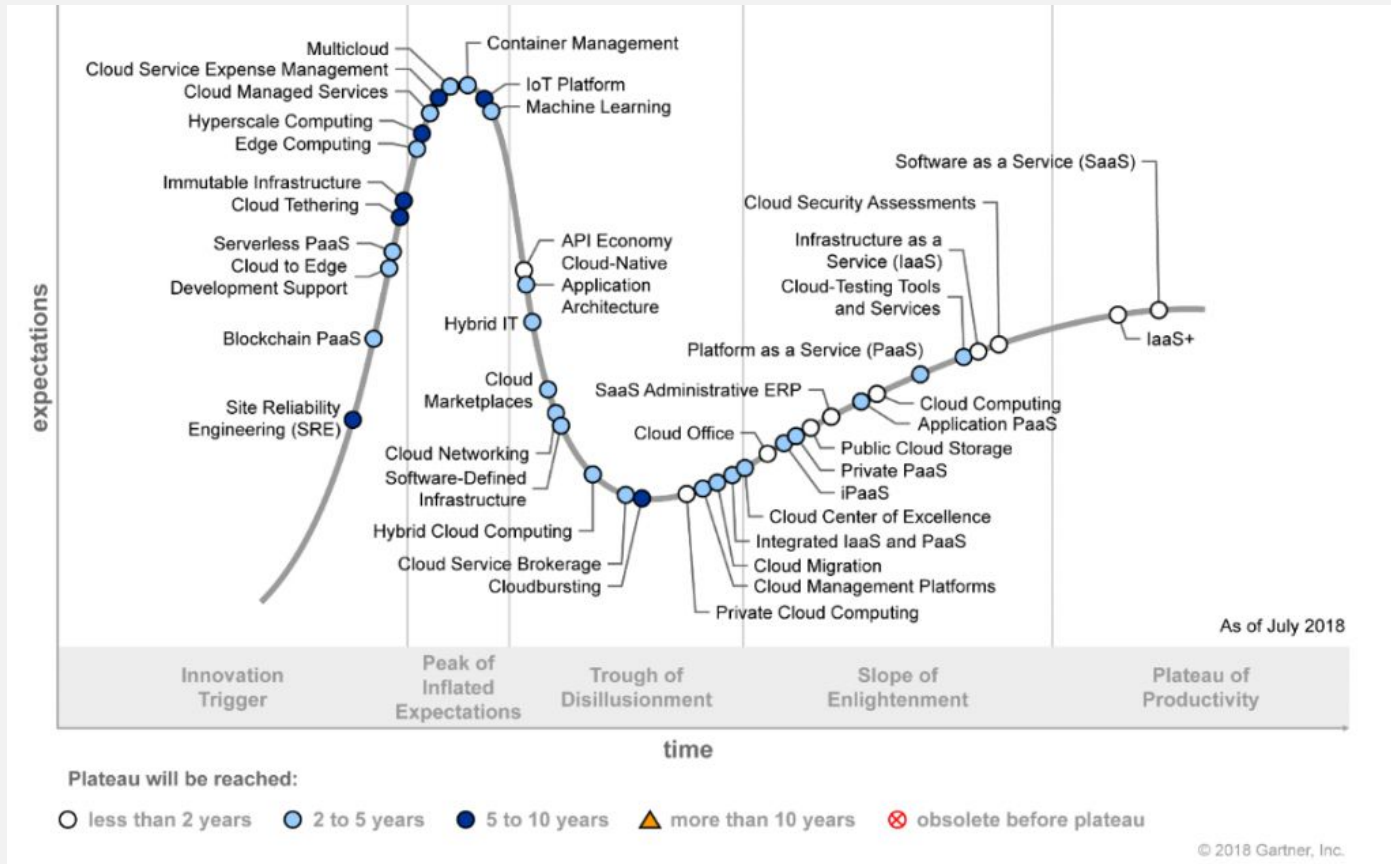
But it returns to the facts

Interesse im zeitlichen Verlauf ?



<https://trends.google.de/trends/explore?date=all&geo=DE&q=cloud%20computing>

Ten years later (2018): cloud computing is a commodity.



Cloud computing is not a surprise, but has been created on the shoulders of giants.



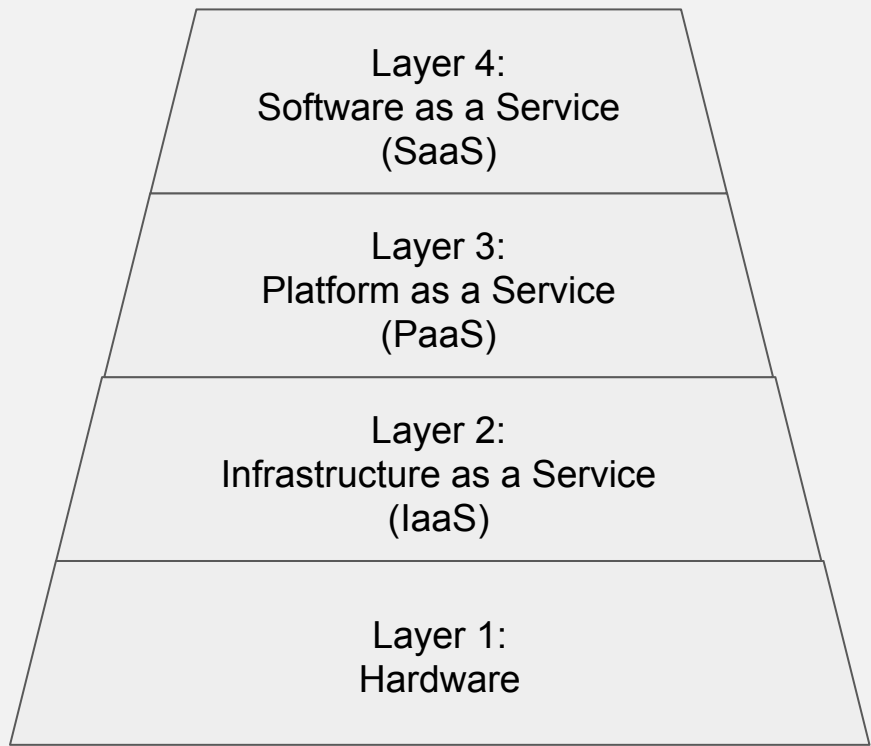
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- **Virtualization** (Virtual machines, hardware virtualization)
- **Commoditization of hardware**
- **Commoditization of the internet** (broadband access, various end devices)
- **Distributed processing** (remote protocols, distributed memory and storage, grid/cluster computing, peer-to-peer)
- **High performance computing** (Parallel algorithms, parallel computers, GPU computing, NoSQL databases)



Cloud Computing

The layered model of cloud computing: From metal to application.



- Customizable software services
- CaaS: Component as a service (e.g. Google Charts)
- BaaS: Backend as a Service
- Transparent Updates

Target Group:
Users

- Platform services
- Application programming interfaces (APIs)
- Abstraction of technical infrastructure

Target group:
Developers

- Elasticity
- Virtual resourcepools
- Technical infrastructure: Machines, Servers (DNS, DHCP, LB, NAS, ...)

Target group:
Operations

- Computers
- Network
- Storage

https://www.youtube.com/watch?v=M988_fsOSWo

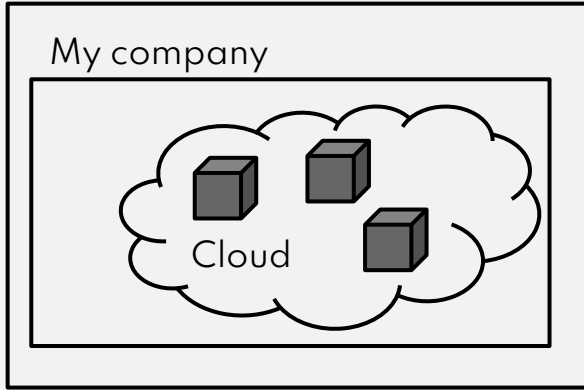


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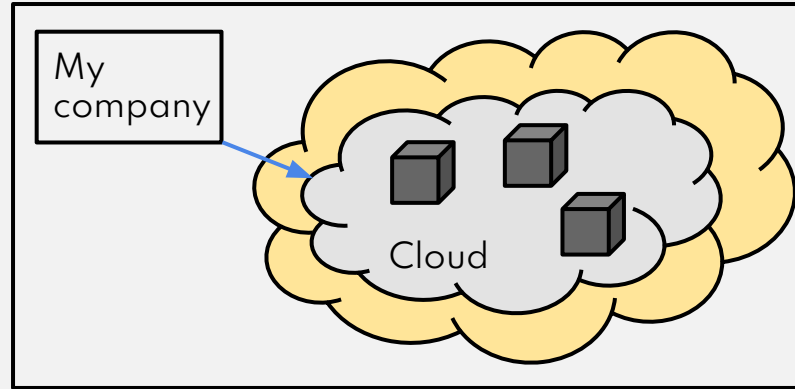
Types of Clouds

Public and private clouds.

Private Cloud

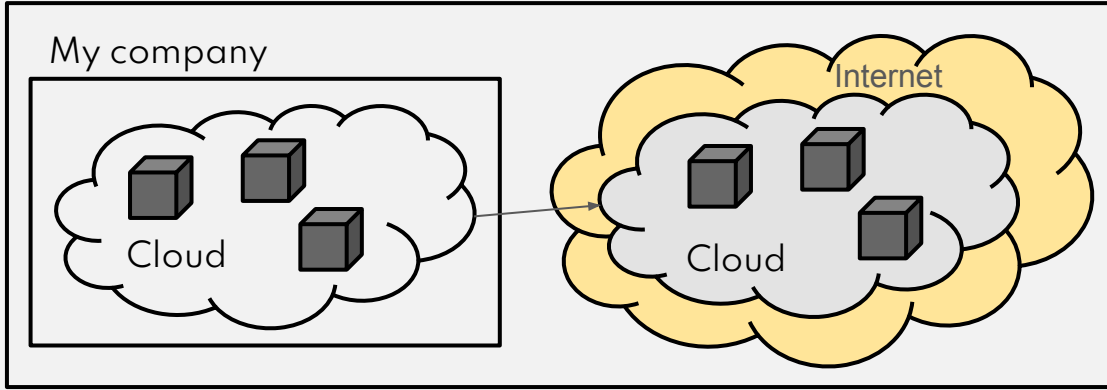


Public Cloud

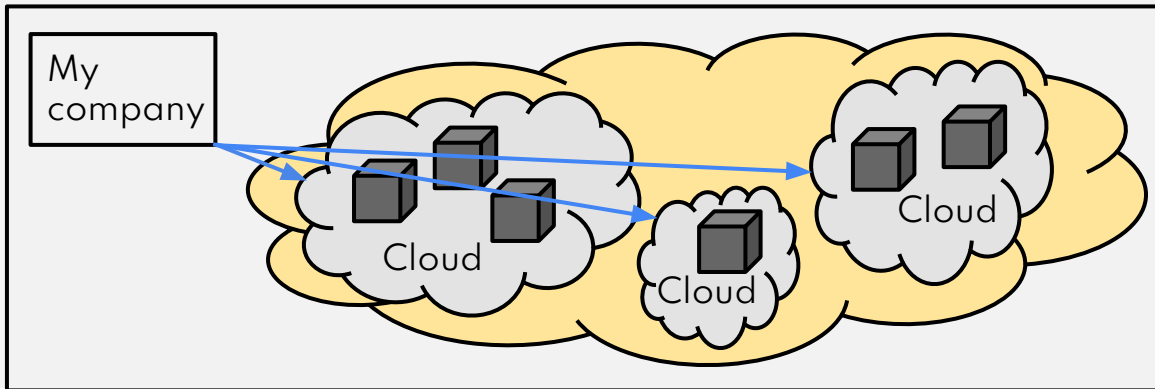


Hybrid and multi Clouds.

Hybrid Cloud



Multi Cloud



Cloud Computing and security.

- Cloud computing currently has a security problem in the public eye and thus a problem with acceptance.
- Legal requirements and certifications for data protection and IT security:
 - BDSG / EU-DSGVO: Bundesdatenschutzgesetz, EU Data Protection
 - TKG: Telekommunikationsgesetz
 - TMG: Telemediengesetz
 - BSI: Bundesamt für Sicherheit in der Informationstechnologie (BSI Grundschutz)
 - ISO 27001: Information security management
 - ISO 18028: IT-Sicherheitsverfahren
- Additional sources:
 - Legal requirements for cloud computing, IT summit
http://www.eurocloud.de/wp-content/blogs.dir/5/files/anford_recht_beicloudcomputing_v1.pdf
- Cloud Security Alliance: <https://cloudsecurityalliance.org>

20.07.2009

E-Reader Kindle

SPIEGEL ONLINE

Amazon löscht digitale Exemplare von "1984"

Amazon löscht Eigentum seiner Kunden: Ausgerechnet die Orwell-Bücher "1984" und "Farm der Tiere" verschwanden aus dem Speicher von Kindle-Lesegeräten, obwohl deren Besitzer sie gekauft und bezahlt hatten. Ein Lehrstück über Macht und Rechte im Zeitalter totaler Vernetzung.



Cloud-Computing

Die Wolke des Grauens

Sie soll der IT-Branche Milliardenerelöse bescheren. Doch viele Kunden schlagen sich mit Sicherheitsbedenken herum und zögern mit dem Umzug in die Datenwolke. Der Aufbau von Vertrauen dauert - und internationale Gütesiegel fehlen. von Annika Graf, Hamburg

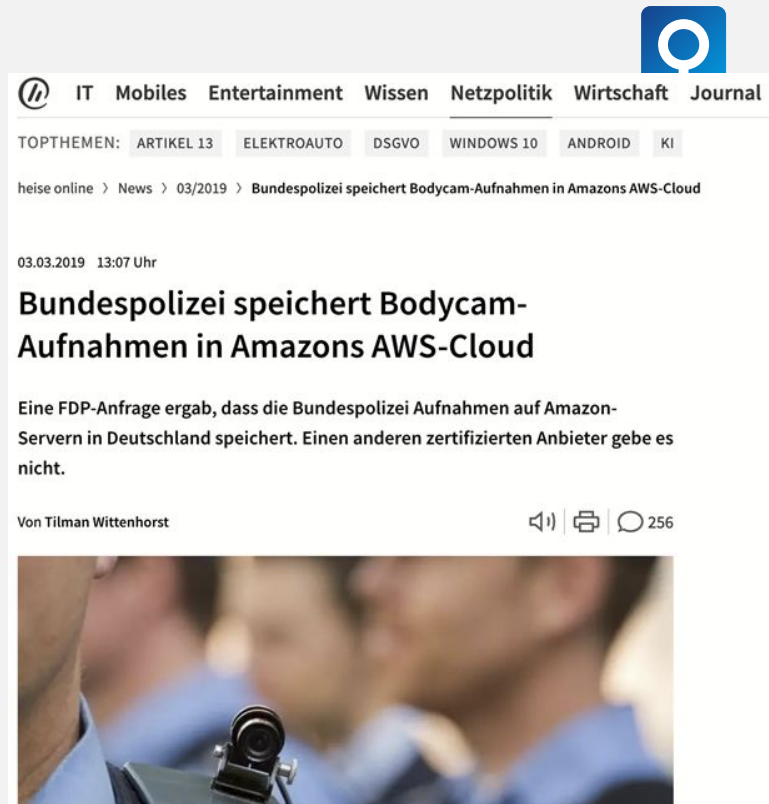
5.3.2012,

<http://www.ftd.de/it-medien/medien-internet/cloud-computing-die-wolke-des-grauens/7003428.html>

Cloud Computing and security.

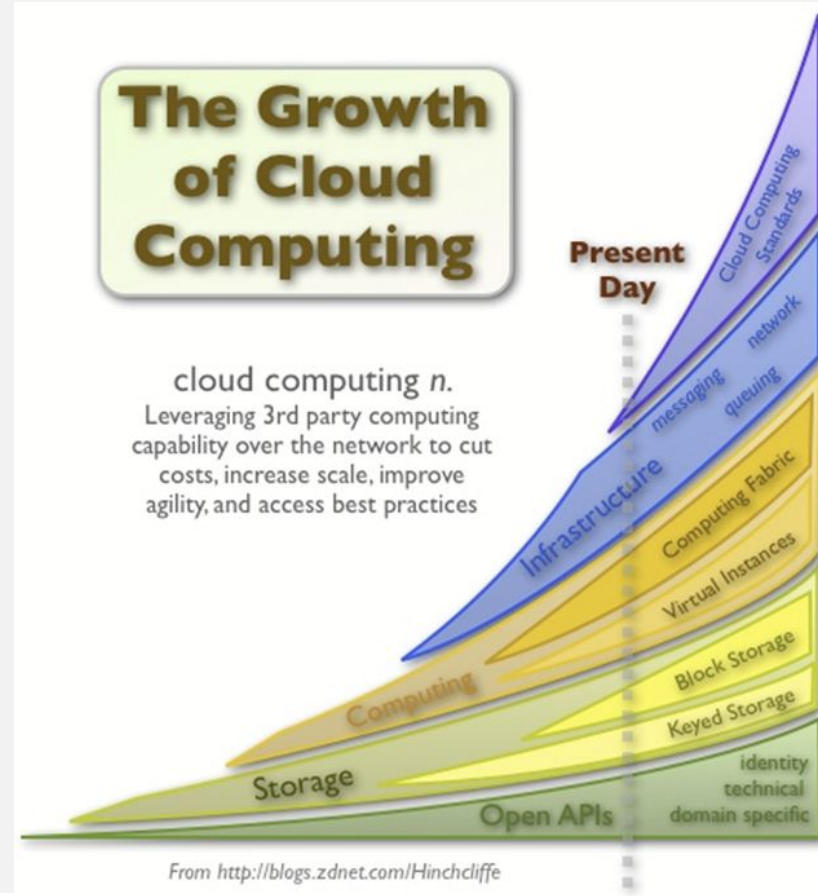
Example: Storing videos in AWS (Amazon Web Services)

- Access by government authorities to data stored in clouds of domestic companies abroad has long been a controversial issue.
- CLOUD Act (Clarifying Lawful Overseas Use of Data Act)
- Foreign authorities should also have access to US servers under the same conditions.
- Who owns the data and who has access to it?
- In this case, the data is stored in encrypted form on AWS.



What can we expect?

- Sharply falling prices for cloud computing.
- Standardization and open interfaces.
- Commodification of the cloud
- Technological expansion
 - Infrastructure capacities
 - Platform services
 - Processing patterns



OLD MAN YELLS AT CLOUD



Old man Abraham Lincoln
yells at cloud

Had given up
not, I don't know if
just had a few
in a few days
d' hdy for the job
so, some of the
probably not a
to be a
to be a
to be a

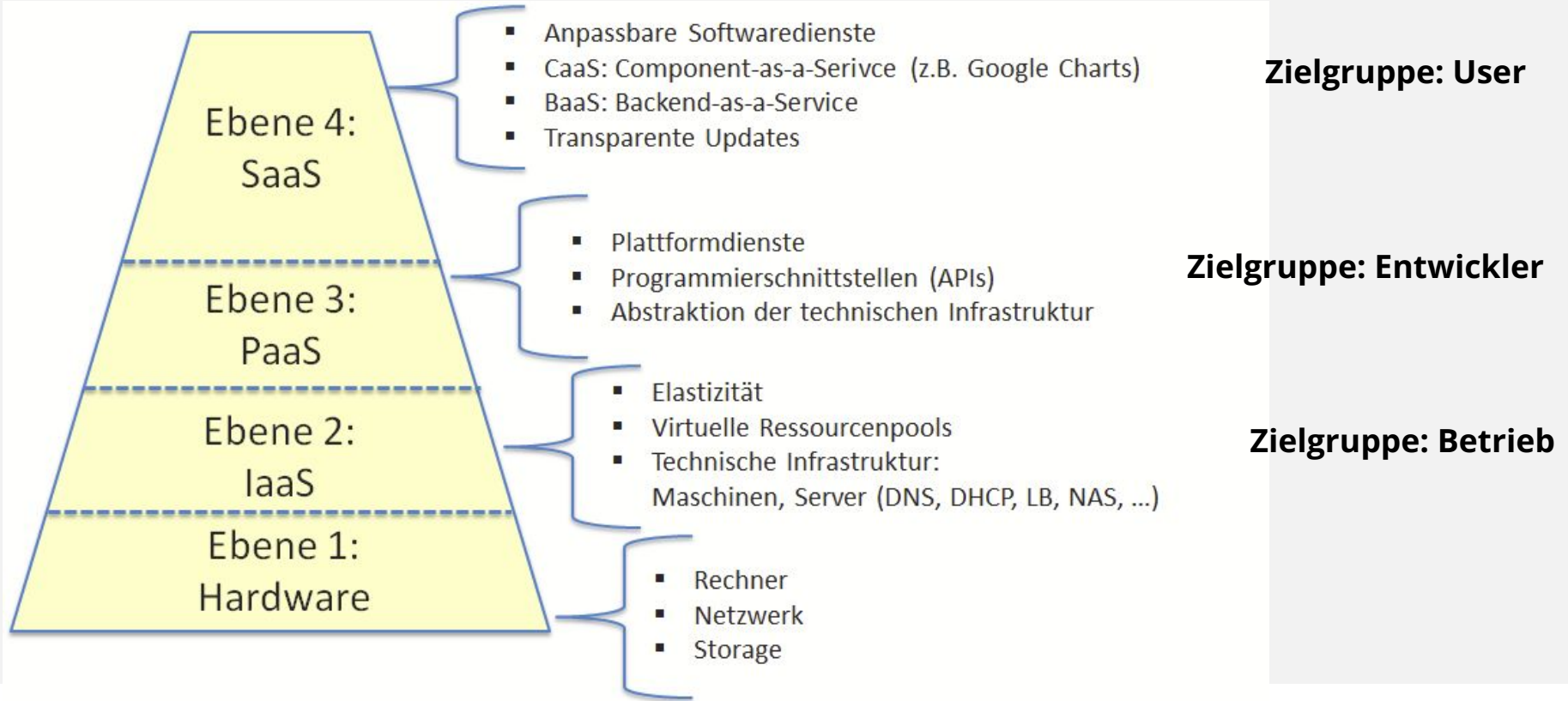
to be a
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?aaS-Quiz!

Das Schichtenmodell des Cloud Computing: Vom Blech zur Anwendung.



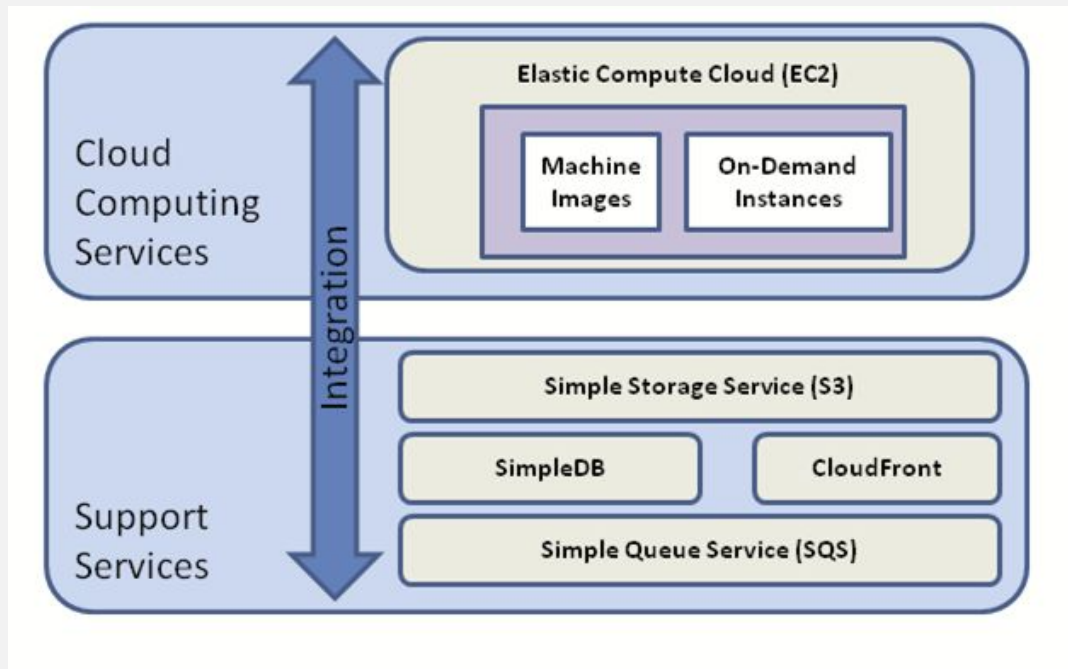
Amazon Kindle

Introducing kindle cloud reader



Read books instantly in your browser
with our free Kindle reading app

Amazon EC2



Google App Engine



App Engine

Dropbox

Was ist Dropbox?

"Deine Projekte, immer und überall."

Dropbox ist ein kostenloser Service, mit dem du deine Fotos, Dokumente und Videos immer zur Hand hast. Das bedeutet, dass jede Datei in deiner Dropbox automatisch auf all deinen Computern, Telefonen und sogar der Dropbox-Website gespeichert wird.

Auch die Freigabe von Dateien ist mit Dropbox ein Kinderspiel - für Studenten, Eltern, Großeltern oder im Büro. Und falls du einmal versehentlich deinen Kaffee über deinen Laptop schüttetest: nur keine Panik! Dropbox ist dein Retter in der Not und sorgt dafür, dass deine Dateien niemals verloren gehen.



Microsoft Azure.

Microsoft Azure Ressourcengruppen

Suchen Sie nach Ressourcen, Diensten und Dokumenten

Ressourcengruppen
R8cdef31-a31e-4b4a-93e4-5f571e91255a

+ Hinzufügen Tags zuweisen Spalten Aktualisieren

Das Verzeichnis "R8cdef31-a31e-4b4a-93e4-5f571e91255a" enthält keine Abonnements.

Nach Name filtern...

0 Elemente

NAME	ABONNEMENT
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Keine Ressourcengruppen zum Anzeigen.

Ändern Sie Ihre Filter, wenn die gesuchten Elemente nicht angezeigt werden. Weitere Informationen

Ressourcengruppe erstellen

Spotify



Cloud Testing.



Sauce Labs OnDemand

Run your Selenium tests in our cloud.
Save Yourself a QA Lab.

[Subscribe Now](#) [Run a Test](#)

13,249,014 - Tests Served

The banner features the Sauce Labs logo in the top left, followed by the text 'OnDemand'. Below this is a tagline and two buttons. On the right is a graphic of a cloud containing several interlocking gears with different colored icons. At the bottom right, it states '13,249,014 - Tests Served'.



WHAT IS BLAZEMETER?

BlazeMeter is a 100% JMeter-compatible, self-service load testing cloud. Instantly generate massive stress tests with comprehensive reporting and analysis tools.

Salesforce



Literature

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<http://thenewstack.io>

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